

From Gamma Functions to Statistical Inference

Beta, chi-square, Student's t, and F distributions



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Gamma function



Gamma distribution

$$\Gamma(a) = \int_0^{\infty} x^{a-1} e^{-x} dx$$

$$X \sim \text{Gamma}(\alpha, \beta)$$

rate parameter β

Gamma function $\longrightarrow$ Gamma distribution $\longrightarrow$ MGF identifications

The Gamma function: factorials beyond the integers

Definition

For $a > 0$,

$$\Gamma(a) = \int_0^{\infty} x^{a-1} e^{-x} dx.$$

- The integrand is positive, and the integral is finite for every $a > 0$.
- Integration by parts gives the fundamental recurrence:

$$\Gamma(a+1) = a \Gamma(a).$$

- Since $\Gamma(1) = 1$, for integers $n \geq 1$,

$$\Gamma(n) = (n-1)!.$$

A special value

$\Gamma(1/2)$

Put $x = u^2$:

$$\Gamma\left(\frac{1}{2}\right) = 2 \int_0^{\infty} e^{-u^2} du = \sqrt{\pi}.$$

Hence

$$\Gamma\left(n + \frac{1}{2}\right) = \frac{(2n)!}{4^n n!} \sqrt{\pi}.$$

The Gamma distribution (rate parameterization)

Definition

A positive random variable X has a Gamma distribution with shape $\alpha > 0$ and rate $\beta > 0$ if

$$X \sim \text{Gamma}(\alpha, \beta), \quad f_X(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0.$$

- **Shape** α controls the behavior near zero and the overall skew.
- **Rate** β is the reciprocal scale: larger β concentrates mass closer to zero.
- Beware: some books use a scale $\theta = 1/\beta$ instead.

MGF, mean, and variance

MGF of $X \sim \text{Gamma}(\alpha, \beta)$

For $t < \beta$,

$$M_X(t) = \mathbb{E}(e^{tX}) = \left(\frac{\beta}{\beta - t} \right)^\alpha.$$

Differentiate at $t = 0$:

$$\mathbb{E}(X) = M'_X(0) = \frac{\alpha}{\beta}.$$

Using $\text{Var}(X) = M''_X(0) - [M'_X(0)]^2$:

$$\text{Var}(X) = \frac{\alpha}{\beta^2}.$$

A couple of concrete Gamma calculations

Example 1: an ordinary shape

If $X \sim \text{Gamma}(4, 3)$, then

$$\mathbb{E}(X) = \frac{4}{3}, \quad \text{Var}(X) = \frac{4}{9}, \quad M_X(t) = \left(\frac{3}{3-t} \right)^4.$$

Example 2: a half-shape

If $Y \sim \text{Gamma}(\frac{1}{2}, \frac{1}{2})$, then

$$\mathbb{E}(Y) = 1, \quad \text{Var}(Y) = 2, \quad M_Y(t) = (1 - 2t)^{-1/2}.$$

Three random quantities to keep in mind

- ① Let $Z \sim \mathcal{N}(0, 1)$ and consider $Y = Z^2$.
- ② Let $Z_1, \dots, Z_d$ be independent $\mathcal{N}(0, 1)$ variables and set

$$Q_d = Z_1^2 + \dots + Z_d^2.$$

- ③ Let $E_1, \dots, E_n$ be iid $\text{Exp}(\lambda)$ variables and set

$$S_n = E_1 + \dots + E_n.$$

The square of a standard normal

Direct MGF calculation

For $Z \sim \mathcal{N}(0, 1)$ and $t < 1/2$,

$$\begin{aligned} M_{Z^2}(t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(tz^2 - \frac{z^2}{2}\right) dz \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left[-\frac{1-2t}{2}z^2\right] dz = (1-2t)^{-1/2}. \end{aligned}$$

$$Z^2 \sim \text{Gamma}\left(\frac{1}{2}, \frac{1}{2}\right) \qquad \mathbb{E}(Z^2) = 1, \text{ Var}(Z^2) = 2.$$

Chi-square as a Gamma distribution

Definition and identification

If $Z_1, \dots, Z_d$ are independent standard normals, then

$$\chi_d^2 := \sum_{j=1}^d Z_j^2 \sim \text{Gamma}\left(\frac{d}{2}, \frac{1}{2}\right).$$

Mean

$$\mathbb{E}(\chi_d^2) = \frac{d/2}{1/2} = d.$$

Variance

$$\text{Var}(\chi_d^2) = \frac{d/2}{(1/2)^2} = 2d.$$

For example, $\chi_4^2 \sim \text{Gamma}(2, 1/2)$, so its mean is 4 and its variance is 8.

A sum of iid exponential waiting times

Start with one waiting time

If $E \sim \text{Exp}(\lambda)$ in the rate convention, then

$$f_E(x) = \lambda e^{-\lambda x}, \quad x > 0, \quad M_E(t) = \frac{\lambda}{\lambda - t}, \quad t < \lambda.$$

Let $S_n = E_1 + \cdots + E_n$ with iid copies E_i .

$$M_{S_n}(t) = \prod_{i=1}^n M_{E_i}(t) = \left(\frac{\lambda}{\lambda - t} \right)^n.$$

MGF identification

$$S_n \sim \text{Gamma}(n, \lambda), \quad \mathbb{E}(S_n) = \frac{n}{\lambda}, \quad \text{Var}(S_n) = \frac{n}{\lambda^2}.$$

Summary

❶ $\Gamma(a+1) = a\Gamma(a)$ and $\Gamma(n) = (n-1)!$.

❷ In the **rate** parameterization,

$$X \sim \text{Gamma}(\alpha, \beta) \implies M_X(t) = \left(\frac{\beta}{\beta - t} \right)^\alpha, \quad \mathbb{E}(X) = \frac{\alpha}{\beta}, \quad \text{Var}(X) = \frac{\alpha}{\beta^2}.$$

❸ Independent sums multiply MGFs, which explains the closure of the Gamma family under common-rate sums.

❹ The key identifications are

$$Z^2 \sim \text{Gamma}\left(\frac{1}{2}, \frac{1}{2}\right), \quad \chi_d^2 \sim \text{Gamma}\left(\frac{d}{2}, \frac{1}{2}\right), \quad \sum_{i=1}^n E_i \sim \text{Gamma}(n, \lambda).$$

Beta distribution

If $U \sim \text{Beta}(a, b)$, with $a, b > 0$, then

$$f_U(u) = \frac{1}{B(a, b)} u^{a-1} (1-u)^{b-1}, \quad 0 < u < 1,$$

where

$$B(a, b) = \int_0^1 u^{a-1} (1-u)^{b-1} du.$$

Order statistics of a uniform sample

Let $U_1, \dots, U_n \stackrel{\text{iid}}{\sim} \text{Unif}(0, 1)$ and let $U_{(1)} \leq \dots \leq U_{(n)}$ be the order statistics.

$$U_{(k)} \sim \text{Beta}(k, n - k + 1).$$

Order statistics of a uniform sample

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$$U_{(k)} \sim \text{Beta}(k, n - k + 1).$$

For $0 < u < 1$,

$$\begin{aligned} f_{U_{(k)}}(u) &= \frac{n!}{(k-1)!(n-k)!} u^{k-1} (1-u)^{n-k} \\ &= \frac{1}{B(k, n-k+1)} u^{k-1} (1-u)^{n-k}. \end{aligned}$$

The Beta–Gamma relation

For $a, b > 0$,

$$\Gamma(a)\Gamma(b) = \int_0^\infty \int_0^\infty x^{a-1} y^{b-1} e^{-(x+y)} dx dy.$$

Use

$$r = x + y, \quad u = \frac{x}{x + y}, \quad x = ru, \quad y = r(1 - u), \quad dx dy = r dr du.$$

Then

$$\begin{aligned} \Gamma(a)\Gamma(b) &= \left[\int_0^1 u^{a-1} (1 - u)^{b-1} du \right] \left[\int_0^\infty r^{a+b-1} e^{-r} dr \right] \\ &= B(a, b)\Gamma(a + b). \end{aligned}$$

Exercise: Beta-function moments

Let $U \sim \text{Beta}(a, b)$. For $r > -a$ and $s < b$, show that

$$\mathbb{E}[U^r(1-U)^{-s}] = \frac{B(a+r, b-s)}{B(a, b)}.$$

Mean and variance of a Beta random variable

For $U \sim \text{Beta}(a, b)$,

$$\mathbb{E}(U) = \frac{a}{a+b}, \quad \text{Var}(U) = \frac{ab}{(a+b)^2(a+b+1)}.$$

Mean and variance of a Beta random variable

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$$\mathbb{E}(U) = \frac{a}{a+b}, \quad \text{Var}(U) = \frac{ab}{(a+b)^2(a+b+1)}.$$

$$\mathbb{E}(U) = \frac{B(a+1, b)}{B(a, b)} = \frac{a}{a+b},$$

$$\mathbb{E}(U^2) = \frac{B(a+2, b)}{B(a, b)} = \frac{a(a+1)}{(a+b)(a+b+1)},$$

$$\text{Var}(U) = \mathbb{E}(U^2) - [\mathbb{E}(U)]^2 = \frac{ab}{(a+b)^2(a+b+1)}.$$

Gamma–Beta construction

Let X and Y be independent with

$$X \sim \text{Gamma}(a, \lambda), \quad Y \sim \text{Gamma}(b, \lambda).$$

Define

$$S = X + Y, \quad U = \frac{X}{X + Y}.$$

Then

$S \sim \text{Gamma}(a + b, \lambda), \quad U \sim \text{Beta}(a, b), \quad S \perp U.$

Student's t distribution

Let

$$Z \sim N(0, 1), \quad V \sim \chi_d^2, \quad Z \perp V.$$

Define

$$T_d = \frac{Z}{\sqrt{V/d}}.$$

Then

$$\boxed{T_d \sim t_d.}$$

Since $\chi_d^2 \sim \text{Gamma}(\frac{d}{2}, \frac{1}{2})$, this definition uses the distributions already developed.

Deriving the t_d density: CDF and differentiation

① For $t \in \mathbb{R}$,

$$F_{T_d}(t) = \Pr\left(Z \leq t\sqrt{V/d}\right) = \int_0^\infty \Phi\left(t\sqrt{v/d}\right) f_V(v) dv.$$

② Differentiate under the integral sign:

$$f_{T_d}(t) = \int_0^\infty \phi\left(t\sqrt{v/d}\right) \sqrt{v/d} f_V(v) dv.$$

Deriving the t_d density: substitution and result

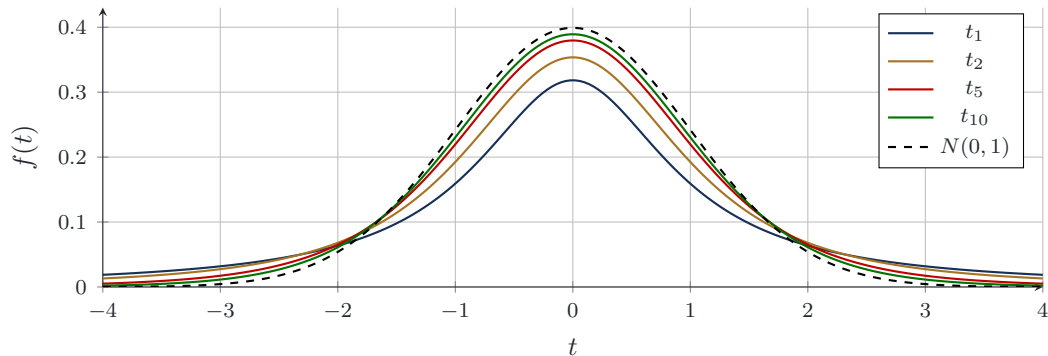
- ③ Insert the normal and chi-square densities:

$$f_{T_d}(t) = \frac{2^{-d/2}}{\sqrt{2\pi d} \Gamma(d/2)} \int_0^\infty v^{(d+1)/2-1} e^{-\frac{1}{2}(1+t^2/d)v} dv.$$

- ④ Set $w = \frac{1}{2}(1+t^2/d)v$ and use the Gamma integral:

$$f_{T_d}(t) = \frac{\Gamma((d+1)/2)}{\sqrt{d\pi} \Gamma(d/2)} \left(1 + \frac{t^2}{d}\right)^{-(d+1)/2}$$

Student's t densities



Curves are computed from the displayed density formula; as d increases, t_d approaches $N(0,1)$.

Cochran's theorem: normal-sample form

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$, with

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Then

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1), \quad \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2,$$

and

$\bar{X}$ and S^2 are independent.

Confidence interval for a normal mean

Cochran's theorem gives

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

Therefore,

$$\Pr\left(-t_{1-\alpha/2, n-1} \leq \frac{\bar{X} - \mu}{S/\sqrt{n}} \leq t_{1-\alpha/2, n-1}\right) = 1 - \alpha.$$

Rearranging gives

$$\boxed{\mu \in \bar{X} \pm t_{1-\alpha/2, n-1} \frac{S}{\sqrt{n}}}.$$

The F distribution

Let U and V be independent with

$$U \sim \chi_{d_1}^2, \quad V \sim \chi_{d_2}^2.$$

Then

$$F = \frac{U/d_1}{V/d_2} \sim F_{d_1, d_2}.$$

Since $U/(U+V) \sim \text{Beta}(d_1/2, d_2/2)$,

$$F = \frac{d_2}{d_1} \frac{U/(U+V)}{1 - U/(U+V)}.$$

In particular,

$$T_d^2 \sim F_{1, d}.$$

Mean and variance of an F random variable

If $F \sim F_{d_1, d_2}$, then

$$\mathbb{E}(F) = \frac{d_2}{d_2 - 2}, \quad d_2 > 2,$$

and

$$\text{Var}(F) = \frac{2d_2^2(d_1 + d_2 - 2)}{d_1(d_2 - 2)^2(d_2 - 4)}, \quad d_2 > 4.$$

$$\text{Use } F = \frac{d_2}{d_1} \frac{U}{1 - U}, \quad U \sim \text{Beta}\left(\frac{d_1}{2}, \frac{d_2}{2}\right).$$

Confidence intervals from chi-square and Gamma pivots

Why these intervals matter

- **Manufacturing:** how variable are the fill volumes from a production line?
- **Calibration:** what is the uncertainty in the mean measurement error?
- **Service systems:** what is the average waiting time or event rate?
- **Process comparison:** how does the variability of one process compare with another?

Data $\longrightarrow$ pivot $\longrightarrow$ confidence interval

How the reference distributions appear

Statistical question	Pivotal quantity	Law
Normal mean, σ known	$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$	$N(0, 1)$
Normal variance, μ known	$\frac{\sum (X_i - \mu)^2}{\sigma^2}$	χ_n^2
Normal mean, σ unknown	$\frac{\bar{X} - \mu}{S/\sqrt{n}}$	t_{n-1}
Ratio of two normal variances	$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}$	F_{n_1-1, n_2-1}
Exponential rate λ	$2\lambda \sum E_i$	χ_{2n}^2

Variance confidence interval: mean known

Suppose $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$ and μ is known. Then

$$\frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu)^2 \sim \chi_n^2.$$

If $\chi_{\nu,p}^2$ is the p -quantile of χ_ν^2 , then

$$\boxed{\frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi_{n,1-\alpha/2}^2} \leq \sigma^2 \leq \frac{\sum_{i=1}^n (X_i - \mu)^2}{\chi_{n,\alpha/2}^2}}.$$

Worked example: variance, mean known

A calibration instrument has known mean error $\mu = 0$. Ten measured errors are

$$2, -1, 3, -2, 1, 0, -3, 2, -1, 1.$$

Construct a 95% confidence interval for σ^2 .

Worked example: variance, mean known

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$$2, -1, 3, -2, 1, 0, -3, 2, -1, 1.$$

Construct a 95% confidence interval for σ^2 .

$$\sum x_i^2 = 34, \quad \chi_{10,.975}^2 = 20.483, \quad \chi_{10,.025}^2 = 3.940.$$

$$\sigma^2 \in \left(\frac{34}{20.483}, \frac{34}{3.940} \right) = (1.66, 8.63).$$

Variance confidence interval: mean unknown

For $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2.$$

Hence

$$\boxed{\frac{(n-1)S^2}{\chi_{n-1, 1-\alpha/2}^2} \leq \sigma^2 \leq \frac{(n-1)S^2}{\chi_{n-1, \alpha/2}^2}}.$$

Worked example: variance, mean unknown

A bottling machine produces the following fill volumes, in mL:

498, 501, 500, 502, 499, 503, 497, 500.

Assuming normality, construct a 95% confidence interval for σ^2 .

Worked example: variance, mean unknown

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498, 501, 500, 502, 499, 503, 497, 500.

Assuming normality, construct a 95% confidence interval for σ^2 .

$$n = 8, \quad \bar{x} = 500, \quad S^2 = 4, \quad (n - 1)S^2 = 28.$$

$$\sigma^2 \in \left(\frac{28}{\chi_{7,.975}^2}, \frac{28}{\chi_{7,.025}^2} \right) = \left(\frac{28}{16.013}, \frac{28}{1.690} \right) = (1.75, 16.57) \text{ mL}^2.$$

Confidence interval for an exponential rate

If $E_1, \dots, E_n \stackrel{\text{iid}}{\sim} \text{Exp}(\lambda)$ and $S = \sum_{i=1}^n E_i$, then

$$S \sim \text{Gamma}(n, \lambda), \quad 2\lambda S \sim \chi_{2n}^2.$$

Therefore, with $\theta = 1/\lambda$,

$$\lambda \in \left(\frac{\chi_{2n, \alpha/2}^2}{2S}, \frac{\chi_{2n, 1-\alpha/2}^2}{2S} \right)$$

and

$$\theta \in \left(\frac{2S}{\chi_{2n, 1-\alpha/2}^2}, \frac{2S}{\chi_{2n, \alpha/2}^2} \right).$$

Worked example: exponential waiting times

The time gaps between help-desk calls, in minutes, are

4, 7, 3, 6, 5, 9.

Assuming iid exponential waiting times, construct a 95% confidence interval for the rate λ and mean waiting time $\theta = 1/\lambda$.

Worked example: exponential waiting times

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$$4, 7, 3, 6, 5, 9.$$

Assuming iid exponential waiting times, construct a 95% confidence interval for the rate λ and mean waiting time $\theta = 1/\lambda$.

$$n = 6, \quad S = 34, \quad \chi_{12,.025}^2 = 4.404, \quad \chi_{12,.975}^2 = 23.337.$$

$$\lambda \in (0.0648, 0.3432) \text{ per minute}, \quad \theta \in (2.914, 15.441) \text{ minutes}.$$

Confidence interval for a ratio of variances

For two independent normal samples,

$$R = \frac{\sigma_1^2}{\sigma_2^2}, \quad \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} = \frac{S_1^2/S_2^2}{R} \sim F_{n_1-1, n_2-1}.$$

Thus a $100(1 - \alpha)\%$ confidence interval for R is

$$\left[\frac{S_1^2/S_2^2}{F_{1-\alpha/2; n_1-1, n_2-1}}, \frac{S_1^2/S_2^2}{F_{\alpha/2; n_1-1, n_2-1}} \right].$$

Worked example: comparing process precision

Two filling machines produce bottle volumes, in mL:

Machine A	100	102	98	101	99	103	97	100
Machine B	98	100	101	99	102	99	100	101

Assuming independent normal measurements, construct a 95% confidence interval for σ_A^2/σ_B^2 .

Worked example: comparing process precision

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Machine A	100	102	98	101	99	103	97	100
Machine B	98	100	101	99	102	99	100	101

Assuming independent normal measurements, construct a 95% confidence interval for σ_A^2/σ_B^2 .

$$S_A^2 = 4, \quad S_B^2 = \frac{12}{7}, \quad \frac{S_A^2}{S_B^2} = \frac{7}{3},$$

$$F_{7,.975} = 4.995, \quad F_{7,.025} = 0.200, \quad \boxed{\frac{\sigma_A^2}{\sigma_B^2} \in \left(\frac{7/3}{4.995}, \frac{7/3}{0.200} \right) = (0.47, 11.66)}.$$

Practice: identify the pivot

For each situation:

- ① identify the parameter of interest;
- ② choose a pivot;
- ③ identify its reference distribution and degrees of freedom;
- ④ invert the probability statement to obtain the interval.

The cases are deliberately mixed.

Practice 1: component lifetime

A normal model is reasonable for component lifetimes. From $n = 16$ components,

$$\bar{x} = 52.4 \text{ hours}, \quad s = 6.8 \text{ hours}.$$

Construct a 95% confidence interval for the mean lifetime μ .

Practice 1: component lifetime

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$$\bar{x} = 52.4 \text{ hours}, \quad s = 6.8 \text{ hours}.$$

Construct a 95% confidence interval for the mean lifetime μ .

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{15}, \quad \mu \in \bar{x} \pm t_{15,.975} \frac{s}{\sqrt{16}}.$$

$$\mu \in 52.4 \pm 2.131 \left(\frac{6.8}{4} \right) = (48.78, 56.02) \text{ hours}.$$

Practice 2: waiting time

Interarrival times at a service desk are modeled as iid exponential. For $n = 8$ observed gaps, the total waiting time is $S = 36$ minutes. Construct a 95% confidence interval for the mean waiting time $\theta = 1/\lambda$.

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$$2\lambda S \sim \chi_{16}^2, \quad \theta \in \left(\frac{2S}{\chi_{16,.975}^2}, \frac{2S}{\chi_{16,.025}^2} \right).$$

$$\theta \in \left(\frac{72}{28.845}, \frac{72}{6.908} \right) = (2.50, 10.42) \text{ minutes.}$$

Practice 3: sensor precision

A sensor has a calibrated mean error of zero. Ten independent errors are normal, and their squared deviations from zero sum to 42. Construct a 95% confidence interval for the error variance σ^2 .

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$$\frac{\sum X_i^2}{\sigma^2} \sim \chi_{10}^2, \quad \sigma^2 \in \left(\frac{42}{\chi_{10,.975}^2}, \frac{42}{\chi_{10,.025}^2} \right).$$

$$\sigma^2 \in \left(\frac{42}{20.483}, \frac{42}{3.940} \right) = (2.05, 10.66) \text{ error units}^2.$$

Practice 4: relative precision

Two independent laboratories report sample variances

$$n_1 = n_2 = 8, \quad S_1^2 = 3.0, \quad S_2^2 = 2.0.$$

Assuming normal measurements, construct a 95% confidence interval for σ_1^2/σ_2^2 .

Practice 4: relative precision

Two independent laboratories report sample variances

$$n_1 = n_2 = 8, \quad S_1^2 = 3.0, \quad S_2^2 = 2.0.$$

Assuming normal measurements, construct a 95% confidence interval for σ_1^2/σ_2^2 .

$$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F_{7,7}, \quad \frac{S_1^2}{S_2^2} = 1.5.$$

$$\frac{\sigma_1^2}{\sigma_2^2} \in \left(\frac{1.5}{4.995}, \frac{1.5}{0.200} \right) = (0.30, 7.50).$$

Practice 5: known measurement error

A measurement process has known standard deviation $\sigma = 4$ units. A sample of $n = 25$ measurements has mean $\bar{x} = 73.2$ units. Construct a 95% confidence interval for the population mean μ .

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$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1), \quad \mu \in \bar{x} \pm z_{.975} \frac{\sigma}{\sqrt{n}}.$$

$$\mu \in 73.2 \pm 1.96 \left(\frac{4}{5} \right) = (71.63, 74.77) \text{ units.}$$

Bayes' theorem: the update rule

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For events A and B with $P(B) > 0$,

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}.$$

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For events A and B with $P(B) > 0$,

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}.$$

$$\text{posterior} = \frac{\text{evidence under } A \times \text{prior}}{\text{total evidence}}.$$

A simple Bayesian example

Suppose 1% of devices have a fault. A diagnostic test is positive with probability 0.90 when a fault is present and 0.05 otherwise.

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$$P(F) = 0.01, \quad P(+ \mid F) = 0.90, \quad P(+ \mid F^c) = 0.05.$$

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$$P(+) = 0.90(0.01) + 0.05(0.99) = 0.0545.$$

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Let F denote the event of a fault and $+$ denote a positive test:

$$P(F) = 0.01, \quad P(+ | F) = 0.90, \quad P(+ | F^c) = 0.05.$$

$$P(+) = 0.90(0.01) + 0.05(0.99) = 0.0545.$$

$$P(F | +) = \frac{0.90(0.01)}{0.0545} \approx 0.165.$$

A positive test changes, but does not determine, our uncertainty.

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- New observations should be incorporated without restarting the analysis.

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Bayesian philosophy

Probability describes uncertainty about the unknown quantity; Bayes' theorem updates that uncertainty when evidence arrives.

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$$\boxed{\pi(\theta \mid x) \propto L(\theta; x) \pi(\theta).}$$

The likelihood is already familiar: Bayes' theorem tells us how to use it to update a distribution for θ .

Sequential evidence and conjugacy

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same algebraic form $\implies$ simple hyperparameter update.

This makes repeated updating transparent.

Bernoulli data: Beta prior

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The likelihood is

$$L(p; x) = p^s(1-p)^f.$$

The Beta family is conjugate for Bernoulli data.

$$\pi(p \mid x) \propto p^s (1 - p)^f p^{a-1} (1 - p)^{b-1}.$$

Bernoulli–Beta update

$$\pi(p \mid x) \propto p^s (1 - p)^f p^{a-1} (1 - p)^{b-1}.$$

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Data add successes to the first shape parameter and failures to the second.

$$p \sim \text{Beta}(2, 2), \quad s = 4, \quad f = 2 \quad \implies \quad p \mid x \sim \text{Beta}(6, 4).$$

Poisson data: Gamma prior in rate format

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If $t = \sum x_i$, then

$$L(\lambda; x) \propto \lambda^t e^{-n\lambda}.$$

Thus Gamma is conjugate for Poisson data.

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For counts $(2, 1, 0, 3)$, a $\text{Gamma}(2, 1)$ prior gives

$$t = 6, \quad \lambda \mid x \sim \text{Gamma}(8, 5).$$

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The likelihood depends on the sample through $\bar{X}$:

$$L(\mu; x) \propto \exp\left\{-\frac{n}{2\sigma^2}(\bar{x} - \mu)^2\right\}.$$

The Normal family is conjugate for a Normal mean with known variance.

Normal–Normal update

Define posterior precision and mean by

$$\frac{1}{\tau_n^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}, \quad m_n = \tau_n^2 \left(\frac{m_0}{\tau_0^2} + \frac{n\bar{x}}{\sigma^2} \right).$$

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$$\boxed{\mu \mid x \sim \mathcal{N}(m_n, \tau_n^2).}$$

The posterior mean is a precision-weighted average of prior mean and sample mean.

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With $m_0 = 70$, $\tau_0^2 = 16$, $n = 4$, $\bar{x} = 73$, and $\sigma^2 = 9$,

$$\tau_n^2 = \frac{144}{73} \approx 1.97, \quad m_n = \frac{5302}{73} \approx 72.63.$$

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These are estimates of the unknown parameter, obtained from the posterior; they are not claims that the parameter itself is random in the physical world.

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An equal-tail $(1 - \alpha)$ credible interval is

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Key distinction

A 95% credible interval has posterior probability 0.95 given the data. The posterior is the distribution used to update the interval statement.

Bayesian estimation: the workflow

- ① Specify a prior distribution for the unknown parameter.
- ② Write the likelihood from the sampling model.
- ③ Multiply prior and likelihood to obtain the posterior.
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Conjugacy makes this update algebraic for the Bernoulli–Beta, Poisson–Gamma, and Normal–Normal examples.